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RealGaze: Towards natural gaze estimation ☆

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ABSTRACT

Gaze estimation plays a crucial role in human–computer interaction and intelligent systems. However, existing appearance-based methods typically rely on image normalization using camera intrinsic parameters to mitigate head pose variations and ensure consistent eye size, which restricts their practicality in real-world applications. To address this limitation, we propose RealGaze, an end-to-end gaze estimation framework that operates on real, unnormalized images. To effectively handle head pose variations, we introduce a head pose field branch and a pseudo head pose strategy, enabling the network to exploit head pose information without explicit normalization. Extensive experiments on the MPIIFaceGaze, EyeDiap, and ETH-XGaze datasets demonstrate that RealGaze achieves state-of-the-art performance, yielding improvements of 18.5%, 1.0%, and 4.7%, respectively. Furthermore, integrating our framework into existing gaze estimation methods significantly enhances their performance on real eye images. Additional cross-camera experiments on ETH-XGaze and comparative studies on EyeDiap further confirm the robustness and broad applicability of the proposed approach.

1. Introduction

Gaze estimation plays a pivotal role in the field of computer vision, with its applications spanning across several industries such as [1], virtual reality [2], driver assistance systems [3] and advertising effectiveness analysis [4]. With the increasing demand for natural user interactions and smarter systems, the need for accurate gaze estimation has become more critical than ever. These advancements not only elevate user experiences but also provide vital support for the development of various intelligent applications. For example, in human–computer interaction, intelligent systems can infer a user's intention by analyzing gaze direction, enabling more intuitive and personalized service. Likewise, in driver assistance systems, tracking a driver's gaze can significantly improve road safety by helping prevent accidents, ensuring a more secure driving environment.

With the advancement of deep learning, appearance-based gaze estimation methods have emerged as the dominant approach in the field. These methods predict gaze direction by analyzing eye or face images using a neural network. Prior to being input into the network, these images are typically normalized to eliminate the influence of head pose on gaze estimation, as gaze is influenced not only by eye movement but also by head pose. The raw images captured by cameras contain both eye movement and head pose information, and without normalization, the network would struggle to extract accurate gaze features from the complex data. However, this normalization process relies on camera intrinsic parameters. In many practical scenarios, these parameters may not be accessible, making it impossible to perform the necessary normalization. Moreover, the need for normalization complicates the gaze estimation process. It requires first normalizing the raw images, then using the neural network to predict the gaze direction from the normalized images, and finally converting the predicted gaze back into the real-world gaze using a rotation matrix derived from the initial normalization step. This multi-step process can be cumbersome, especially in deployment settings where obtaining camera parameters may be challenging.

To address this challenge, we propose a novel method called RealGaze, which eliminates the need for normalization and enables end-to-end gaze estimation directly from real eye images (see Fig. 1). The objectives of normalization are twofold: first, to correct head pose, ensuring that the face remains centered in the image, and second, to

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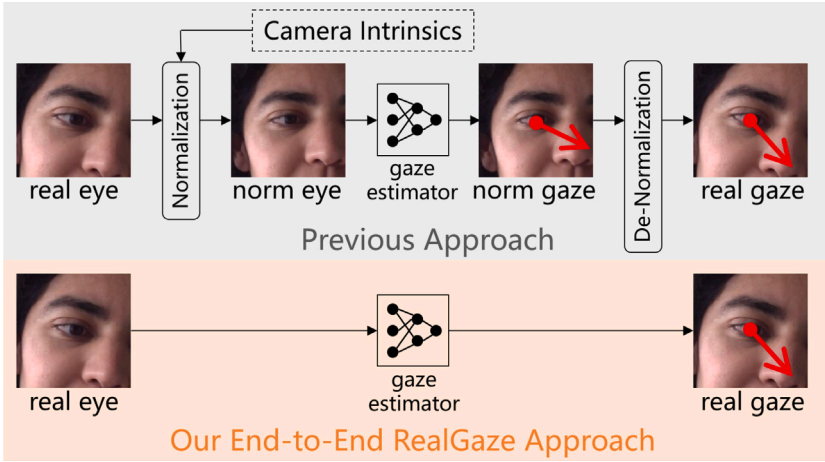

Fig. 1. Comparison between conventional gaze estimation pipelines and the proposed RealGaze method. Conventional approaches rely on face and facial landmark detection, followed by rotation and scaling operations based on camera intrinsics to extract normalized eye or face patches for gaze estimation. In contrast, RealGaze predicts gaze directly from cropped real eye images, requiring only basic face and eye detection as preprocessing and completely eliminating geometric normalization, while maintaining competitive performance.

maintain consistent eye image sizes. RealGaze addresses these objectives in an innovative manner. For the first objective — correcting head pose — we introduce two methods: the head pose field branch and the pseudo head pose strategy. These techniques guide the network to learn head pose information, allowing gaze features to be extracted from real eye images with high accuracy. Specifically, we use the 6DRepnet [5] algorithm to estimate head pose pseudo-labels from real head images. By leveraging the yaw and pitch angles from these pseudo-labels, we generate Gaussian-weighted distribution images, called the head pose field. The pseudo head pose strategy is a simpler approach, where an additional network directly predicts the head pose from the real face image, and this predicted head pose feature map is then incorporated into the gaze estimation network. Since gaze features, head pose field features, and pseudo head pose features are derived from different sources, we design a Multi-Scope Fusion (MSF) module to address the domain discrepancies between these features. This module effectively eliminates the domain discrepancies, allowing the network to extract more accurate gaze features. For the second objective — maintaining consistent eye image sizes — we crop a square region centered at the midpoint between the inner and outer corners of the eye, with the side length being four times the distance between these two corners. This cropped region is then resized to a uniform size, ensuring that the eye images remain consistent in dimension across all inputs, thus facilitating reliable gaze estimation without the need for traditional normalization.

We evaluated the performance of the proposed method on several public datasets, including MPIIFaceGaze [6], EyeDiap [7], and ETH-XGaze [8]. The experimental results demonstrate that our method achieves state-of-the-art performance across all three datasets. Specifically, the error on MPIIFaceGaze from 5.99° to 4.88°, on EyeDiap from 6.76° to 6.69°, and on ETH-XGaze from 4.68° to 4.46°. We also applied our framework to existing gaze estimation approaches, such as Gaze360 [9], XGaze [8], GazeTR [10], and GazePTR [11], and observed substantial performance improvements on real images. Furthermore, we conducted cross-camera experiments using the ETH-XGaze dataset, which showed that our method significantly outperforms normalization-based approaches. To further demonstrate the advantages of RealGaze, we designed an experiment on the EyeDiap dataset, comparing errors under different normalization settings. The results revealed that our method is not only highly robust but also delivers superior performance compared to traditional normalization techniques. Lastly, we performed a visualization analysis of the input and output feature maps from the Multi-Scope Fusion (MSF) module. This analysis helped uncover the underlying reasons for its effectiveness, showcasing how the MSF module facilitates more accurate gaze estimation by overcoming domain discrepancies between different feature sources.

In summary, our contributions are as follows:

1. We propose RealGaze, an end-to-end gaze estimation algorithm specifically designed for real images. This approach minimizes the reliance on image normalization, enabling accurate gaze prediction directly from raw images.
2. We introduce a head pose field branch, where the pose field is generated from pseudo head pose angles. This branch effectively incorporates head pose direction information into the model, allowing the network to better account for head orientation.
3. We design a pseudo head pose strategy to assist multi-scope feature fusion achieved through the proposed MSF module. This strategy uses the pseudo head pose angles to implicitly embed head pose cue into the fused feature, mitigating the loss of head pose awareness that arises under normalization-free conditions.

2. Related work

2.1. Appearance-based gaze estimation

Appearance-based gaze estimation methods have gained prominence due to their capability to directly learn gaze directions from image data without requiring elaborate geometric models. These methods primarily leverage deep learning techniques to extract features from eye or face images and estimate gaze direction.

Kellnhofer et al. [9] proposed a temporal gaze estimation method capable of achieving 360-degree omnidirectional gaze estimation. Zhao et al. [12] introduced a calibration-free personalized gaze estimation network. Cheng et al. [10] combined Transformer models with CNNs to achieve an end-to-end full-face gaze estimation method. Building on this, they drew inspiration from the feature pyramid approach, incorporating a feature pyramid loss function and optimizing the normalized gaze estimation algorithm using real images [11]. Wang et al. [13] proposed a style transfer method that significantly reduces gaze estimation errors for specific subjects. Abdelrahman et al. [14] decomposed the gaze estimation task into regression and classification tasks and optimized gaze estimation errors through multi-task joint training. Jia et al. [15] introduced a frequency-spatial interaction network that explicitly models complementary global contextual information in the frequency domain and local facial details in the spatial domain via

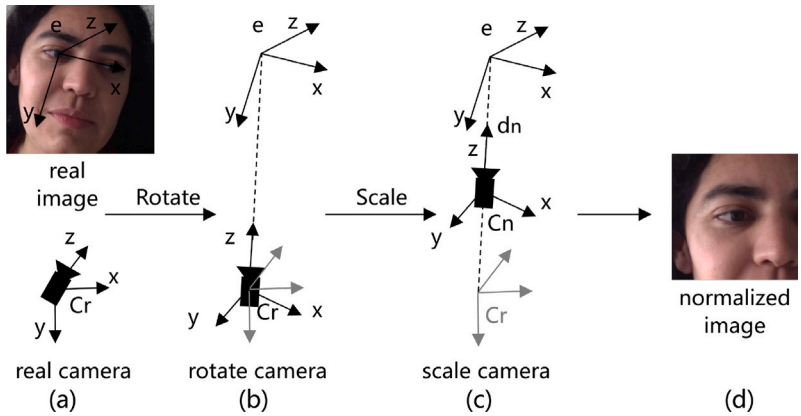

Fig. 2. Eye image normalization process. (a) An initial arbitrary alignment between the head pose coordinate system and the camera coordinate system. (b) The camera coordinate system is rotated using a rotation matrix to achieve canonical alignment. (c) The camera coordinate system is scaled using a scaling matrix to enforce a fixed eye-camera distance. (d) The resulting normalized eye image captured by a virtual normalized camera [18].

dual-branch synergistic modules and a dual-domain interaction block for effective feature fusion. Zhang et al. [16] proposed an end-to-end framework that introduces multi-scale adaptive feature modulation with independent multi-head computation and scale-aware feature fusion to explicitly model global context and inter-scale relationships from facial images alone. Zhang et al. [17] proposed a differential contrastive training strategy that reformulates gaze estimation as a pairwise semantic comparison problem, integrating a CLIP-guided semantic difference-aware branch with a visual appearance-aware network to enhance gaze-related feature learning through image–text alignment.

Although these methods have significantly advanced the field of gaze estimation, most of them, except for Kellnhofer's approach, rely on data normalization. The performance gains they achieve are based on normalization process. For instance, Kellnhofer's method, which operates on the non-normalized Gaze360 dataset, achieves a gaze estimation error of approximately 10 degrees. In contrast, other methods report errors ranging from 4 to 6 degrees on normalized datasets such as MPIIFaceGaze [6], EyeDiap [7], and ETH-XGaze [8]. However, if these datasets were not normalized, whether their performance would remain as strong is questionable.

2.2. Data normalization in gaze estimation

Gaze is influenced by both head movement and eye movement. When the eyes remain stationary, head rotation alters the gaze direction. However, the eye images themselves exhibit little to no visible change. To eliminate the influence of head movement on gaze, many researchers have proposed various image normalization methods. Huang et al. [19] propose a simple normalization method where they first detected the positions of the inner and outer eye corners, aligned the line connecting these corners horizontally, and then cropped the eye region based on the upper and lower eye keypoints. Sugano et al. [20] were the first to propose a method of normalizing images by rotating and scaling the real camera coordinate system. Zhang et al. [18] further refined this method, making normalization a mainstream approach.

The normalization process includes two steps: rotate and scale. As shown in Fig. 2, the rotate step aims to direct the camera towards the eye center, reducing the impact of head pose variations. The scale step ensures that the distance between the camera and the eye remains constant, capturing eye images of the same size and avoiding the problem of varying image sizes due to distance. These two operations produce normalized images that largely eliminate the interference of head pose.

However, the normalization method also has some problems. Firstly, it requires knowledge of the camera's intrinsic parameters, which are not easily obtained in real-world scenarios. Secondly, after normalization, the image undergoes a perspective projection, which means it is no longer a true image. The distribution of normalized images may not be consistent with that of real images. Additionally, after normalization, the predicted gaze direction needs to be inversely rotated to obtain the actual gaze direction. Balim et al. [21] recognized these problems and proposed a frame-to-gaze end-to-end gaze estimation method that directly predicts gaze from the real camera frames. But this method only works when there is a single face in the frame. Zhang et al. [22] proposed an end-to-end method for multi-person gaze estimation, but their method is complex and the gaze estimation error is significantly higher compared to normalized gaze estimation methods. Huang et al. [23] derived normalization parameters from the original image resolution and the face bounding box. Although this approach does not rely on the true camera intrinsic parameters, it still requires normalization operations similar to rotation and scaling. Gaze is first predicted in the normalized space and then transformed back to the original space.

2.3. Feature fusion in gaze estimation

Feature fusion is essential for enhancing the precision and robustness of gaze estimation models. Bao et al. [24] recalibrated eye features by employing shift and scale parameters derived from face features. Gideon et al. [25] disentangled image features through feature swapping in multi-view videos. Yun et al. [26] introduced a high-frequency attention block to amplify high-frequency details, particularly in the eye regions. Hisadome et al. [27] proposed rotation-constrained feature fusion, utilizing relative camera rotation for extracting and merging features. Chen et al. [28] fused disentangled global gaze-relevant features with local eye features through a multi-scale global–local attention mechanism. Li et al. [29] employed a multi-stream architecture to extract eye-only features and eye-face fused features in parallel and then adaptively fused these complementary streams to enhance gaze-relevant representation learning. Wu et al. [30] proposed a dually supervised mixed-attention network that fuses face and ocular features via mixed attention units, enabling effective integration of gaze-relevant and head-pose-related information. However, these methods are typically applied to eye gaze features or facial gaze features, which belong to the same type of features. Their effectiveness on other types of features, such as head pose fields, remains to be studied.

3. Methods

3.1. Preliminary

Existing gaze estimation methods typically rely on image normalization to reduce the impact of head pose. As shown in Fig. 2, the

normalization process involves two steps: (1) Rotation: The camera is rotated around the axis connecting the camera and eye origins until its x-axis aligns with the eye coordinate system's x-axis, ensuring the face directly faces the camera. (2) Scaling: The camera is moved along the line connecting the camera and eye origins to maintain a consistent distance. The image size is then adjusted to ensure uniformity, preventing size variations based on distance.

Although this normalization method can eliminate the interference of head pose, allowing the network to focus more on changes in the eye region, the process is complex and requires knowledge of the camera's intrinsic parameters. This greatly limits the applicability of this method in real-world scenarios.

Compared to normalized images, real images lack the rotation and scaling processes. To compensate for this loss, we introduce additional steps to address these issues.

(1) For rotation, we introduce a head pose field branch and a pseudo head pose strategy to guide head pose information into the network.

(2) For scaling, we crop a square eye region centered at the midpoint between the inner and outer corners of the eye, with a side length four times the distance between the inner and outer corners. This cropped region is then resized to the same size, ensuring that the eye images remain consistent in size.

3.2. Overview

The overall algorithm is illustrated in Fig. 3. Six facial landmarks are first detected using Dlib [31], which includes the four eye corners and the two mouth corners. Based on these landmarks, eye and head images are cropped from the real images. The head images are then fed into 6DRepnet [5] to estimate head pose pseudo-labels, which are used to generate the head pose field according to the head pose direction vectors on the 2D image plane. The cropped eye images, head pose field images, and head images are simultaneously input into the network. Three shallow encoders are employed to extract shallow feature maps from each input, which are subsequently fused by the proposed MSF module to produce a unified feature representation. In parallel, a deep encoder further extracts gaze feature vectors from the eye feature map. The gaze features are concatenated with the fused feature vector to regress the final gaze direction. To ensure reliable head pose representation, an additional deep encoder is introduced to regress head pose from the head feature map, supervised by the pseudo head pose labels during training. Together, these components effectively reduce head pose interference and align features from different input domains, resulting in improved accuracy and robustness in gaze estimation.

3.3. Head pose field branch

We utilize the head pose pseudo-labels generated by 6DRepnet [5] to construct the head pose field. Specifically, the key points of the eyes are first detected in the real image to obtain the 2D coordinates of the eye center. Using this point as the center, we generate a Gaussian distribution map with the same spatial resolution as the original image. The value at each pixel in the head pose field follows a Gaussian distribution centered at the eye location, as defined in Eqs. (1) and (2), where $dist$ represents the distance from any point to the eye center, (x_i, y_i) are arbitrary pixel coordinates, (x_e, y_e) is the eye center, and $field_{i,j}$ is the value at any point on the head pose field. The parameter T controls the magnitude of the distribution. In our experiments, T is empirically set to $\frac{1}{300}$.

$$dist = \sqrt{(x_i - x_e)^2 + (y_i - y_e)^2} \tag{1}$$

$$field_{i,j} = e^{-\frac{(dist \cdot T)^2}{2}} \tag{2}$$

Next, we compute the pseudo head pose direction on the 2D image plane using Eq. (3), where $pitch_{hp}$ and yaw_{hp} denote the pseudo head pose labels predicted by 6DRepnet [5], and $headangle2d$ represents the corresponding 2D head pose angle projected from the 3D head pose. Based on this direction, we retain the region within ±45° to focus on the effective head pose area. Since real camera images typically have unequal width and height, we crop the excess width of the Gaussian distribution map to obtain a square image that preserves the relevant head pose information. This resulting square representation is referred to as the head pose field.

$$headangle2d = tan^{-1} \frac{sin(pitch_{hp})}{cos(pitch_{hp}) sin(-yaw_{hp})} \tag{3}$$

The head pose field can be viewed as a grid-based representation in which each cell encodes local head pose information. By integrating this field into the gaze estimation network, the model is able to implicitly capture head pose cues in a spatially aware manner, making the extracted features robust to variations in head orientation and improving overall gaze estimation performance.

3.4. Pseudo head pose strategy

The pseudo head pose strategy aims to explicitly model and extract head pose features from real head images. This is realized through a dedicated pseudo head pose branch along with a corresponding head pose loss function.

Specifically, we incorporate a pseudo head pose branch into the network that takes the real head image as input and predicts head pose angles. The features learned by this branch provide implicit head pose information that is shared with both the head pose field branch and the gaze estimation branch through the MSF module. Pseudo labels generated by 6DRepnet are used as supervision for this branch, and under the guidance of the head pose angle loss, the network is iteratively optimized to learn discriminative and meaningful head pose features from real head images.

By incorporating both the head pose field and the pseudo head pose branch, our method effectively captures rich head pose information, mitigating the negative impact of head pose variations on gaze estimation.

3.5. Multi scope fusion

The Multi-Scope Fusion (MSF) module is a key component of RealGaze, designed to address the domain discrepancies among different input sources, including eye images, head pose fields, and head images. By aligning and integrating features from these heterogeneous domains, the MSF module enables effective information sharing across branches, ensuring robust feature representation and more accurate gaze estimation.

The detailed workflow of the Multi Scope Fusion (MSF) is illustrated in Fig. 4. Suppose we have two feature maps, $Feature_a$ and $Feature_b$, both of size $H \times W \times C$. The fusion process proceeds as follows:

- Element-wise addition: Compute

$$Feature_c = Feature_a + Feature_b \tag{4}$$

resulting in a feature map $Feature_c$ of size $H \times W \times C$.

- Downsampling: Downsample $Feature_c$ to obtain a dense feature map

$$Feature_d \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times C}. \tag{5}$$

- Self-attention: Apply a self-attention module [32] to $Feature_d$ to capture self-correlated features:

$$Feature_e = \text{SelfAttention}(Feature_d), \quad Feature_e \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times C}. \tag{6}$$

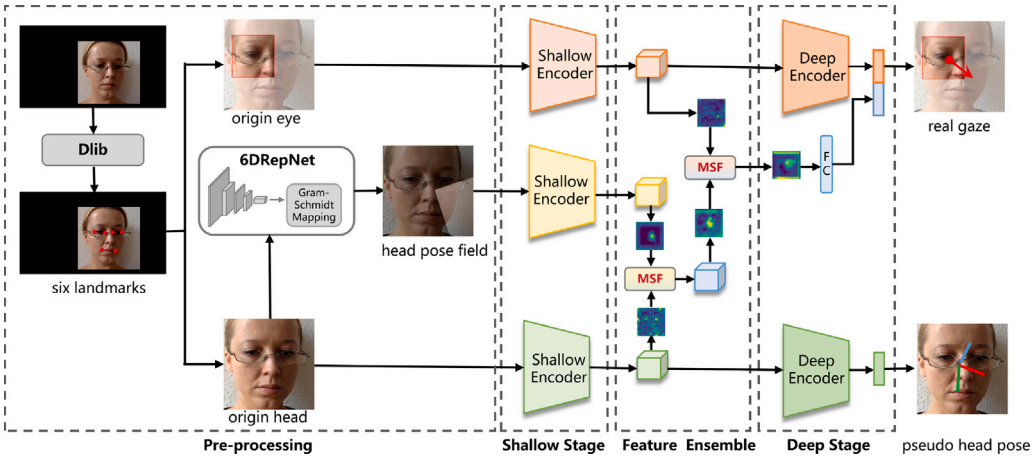

Fig. 3. Pipeline of RealGaze. Facial landmarks are first detected using Dlib [31]. For the head region, the center is defined as the midpoint of six facial keypoints — specifically, the four eye corners and the two mouth corners — and a square region with a side length twice the maximum distance among these keypoints is cropped. For the eye region, a square patch is extracted around the midpoint between the inner and outer eye corners, with a side length four times the inter-corner distance. The head image is then fed into 6DRepnet to estimate pseudo head pose, which is used to generate the head pose field. All cropped images are resized to 160 × 160. The real eye images, head pose fields, and head images are processed by three separate shallow encoders to extract shallow feature maps, which are fused by the proposed Multi-Scope Fusion (MSF) module to obtain a unified representation. The fused feature map is passed through a fully connected layer to produce a fused feature vector, while the shallow eye feature map is further processed by a deep encoder to generate an eye feature vector. The concatenation of these two vectors is used to predict the final gaze direction. Meanwhile, the shallow head pose feature map is fed into a deep encoder to predict pseudo head pose for supervision during training; this branch is removed during inference.

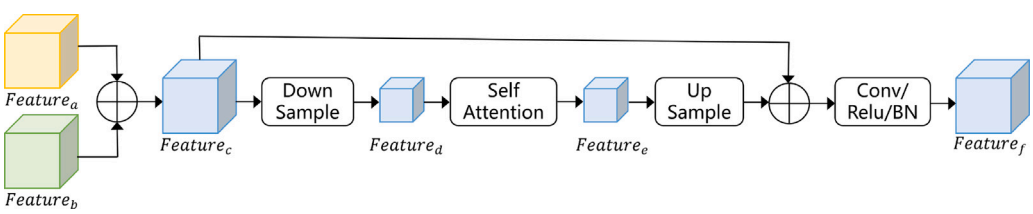

Fig. 4. The architecture of MSF. The MSF module requires that the two input feature maps have the same dimensions. The output feature map produced by MSF also has the same dimensions as the input feature maps.

- Upsampling and shortcut fusion: Upsample $Feature_e$ to the original feature map size and fuse it with the original feature map $Feature_c$ through a shortcut connection.

$$Feature_e^u = \text{Upsample}(Feature_e), \quad Feature_e^u \in \mathbb{R}^{H\times W\times C}. \tag{7}$$

- Convolutional processing: Apply convolution, batch normalization, and ReLU activation to obtain the final fused feature map:

$$Feature_f = \text{ReLU}(\text{BN}(\text{Conv}(Feature_e^u + Feature_c))),$$
$$Feature_f \in \mathbb{R}^{H\times W\times C}. \tag{8}$$

3.6. Loss function

The loss function is composed of two components: a gaze loss and a head pose loss. Both terms are formulated using the L1 loss, as defined in the following equations.

$$\mathcal{L}_{\text{gaze}} = \frac{1}{N}\sum_{i=1}^{N}|\hat{g}_i - g_i| \tag{9}$$

$$\mathcal{L}_{\text{pose}} = \frac{1}{N}\sum_{i=1}^{N}|\hat{p}_i - p_i| \tag{10}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gaze}} + \lambda\mathcal{L}_{\text{pose}} \tag{11}$$

The gaze loss is defined over the yaw and pitch angles of the gaze, while the head pose loss considers the yaw and pitch angles of the head pose, with the roll angle excluded. In our experiments, the weighting factor λ is set to 0.1.

4. Experiments

4.1. Datasets

To evaluate the performance of our algorithm, extensive experiments were conducted on publicly available datasets, MPIIFaceGaze, EyeDiap, and ETH-XGaze.

MPIIFaceGaze: This dataset is widely used for appearance-based gaze estimation and contains 213,659 images collected from 15 participants in everyday environments, featuring unconstrained head poses and normal illumination conditions. Following the dataset protocol, we selected 3000 images per subject using the provided sample list. Facial keypoints were then employed to crop the real eye and head images. For the eye region, the midpoint between the inner and outer eye corners was used as the center, and a square region with a side length four times the distance between these two corners was cropped. For the head region, the center was defined as the midpoint of six facial keypoints, and a square region with a side length twice the maximum distance among these keypoints was extracted. Normalized images were obtained using Cheng's method [33]. Model performance was evaluated using the leave-one-out cross-validation protocol.

EyeDiap: This dataset was collected in an indoor environment using both a Kinect camera and an HD camera, and consists of 237 min of video from 16 subjects, with participants positioned approximately 80–90 cm from the camera. The data processing protocol follows that of MPIIFaceGaze. Specifically, we adopt Cheng's method [33] to select the first 1181 images for each subject and evaluate performance using a 4-fold cross-validation scheme.

Table 1
Comparison with state-of-the-art methods. Eye denotes a single real eye image, Face denotes a single real face image, Two Eyes denotes the left and right real eye images, and Field denotes the head pose field.

Algorithms	Years	Input	MPIIFaceGaze	EyeDiap	ETH-XGaze	GFlops	Params (MB)
Gaze360 [9]	ICCV2019	Eye Face	7.54° 6.18°	7.32° 6.76°	6.44° 4.91°	0.93	11.95
XGaze [8]	ECCV2020	Eye Face	7.34° 6.08°	6.91° 6.81°	6.25° 4.73°	2.11	25.56
GazeTR [10]	ICPR2022	Eye Face	7.36° 6.26°	7.14° 6.81°	6.47° 4.82°	0.94	11.42
GazePTR [11]	CVPR2024	Eye Face	7.36° 6.13°	7.00° 6.97°	6.53° 4.76°	0.96	12.49
CrossGaze [35]	FG2024	TwoEyes+Face	5.99°	6.77°	4.80°	3.29	40.12
GazeSetMerge [34]	ICPR2024	TwoEyes+Face	6.43°	6.82°	4.68°	2.89	80.99
DCGaze [17]	MM2025	Eye Face	8.53° 7.29°	7.19° 7.40°	6.70° 4.91°	7.26	113.49
MAFM-Gaze [16]	Displays2025	Eye Face	8.57° 7.51°	8.10° 7.96°	6.49° 4.68°	3.03	11.63
Baseline(Resnet18)		Eye	7.33°	7.18°	6.48°	0.93	11.69
Ours		Eye+Face+Field	**4.88°**	**6.69°**	**4.46°**	3.12	25.29

ETH-XGaze: This dataset was collected using 18 cameras and comprises a total of 1,083,492 images from 110 subjects. We downloaded the raw dataset and processed the data following a protocol similar to that used for MPIIFaceGaze. As the official test set does not provide ground-truth gaze annotations, we used data from only 80 subjects in the training set. These subjects were split into 56 for training, 12 for validation, and 12 for testing. The model was trained on the sub-training set and evaluated on the sub-test set.

Table 2
Ablation study for head pose field branch (Field), multi-scope fusion (MSF), and pseudo head pose branch (Head).

Field	MSF	Head	MPIIFaceGaze	EyeDiap	ETH-XGaze
×	×	×	7.33°	7.18°	6.48°
✓	×	×	5.76°	6.91°	6.37°
✓	✓	×	5.65°	6.77°	6.17°
✓	✓	✓	**4.88°**	**6.69°**	**4.46°**

4.2. Implementation details

Before being fed into the network, all images are resized to 160 × 160. We adopt ResNet18 as the backbone, using its first and second layers as shallow encoders and the third and fourth layers as deep encoders, and initialize the network with ImageNet-pretrained weights. When applying our method to other algorithms, we use the corresponding model as the backbone for the eye image branch, while keeping the head pose field branch and the pseudo head pose branch unchanged.

During training, the batch size is set to 256 and the model is trained for 24 epochs with an initial learning rate of 1e−3. The learning rate is reduced by half after 18 epochs, and a warmup strategy is applied during the first 5 epochs. The entire framework is implemented in PyTorch and trained on four NVIDIA RTX 3090 GPUs. All methods were trained under identical conditions, including image resolution, batch size, epochs, learning rate, and other parameters.

4.3. Comparison with state-of-the-art methods

Many existing methods focus on face gaze estimation, while our task specifically addresses eye gaze estimation. To ensure a fair comparison, we modified the inputs for the following algorithms — Gaze360 [9], XGaze [8], GazeTR [10], GazePTR [11], DCGaze [17], and MAFM-Gaze [16] — to either single-eye images or face images, with the corresponding outputs being either eye gaze or face gaze.

For all six algorithms — Gaze360, XGaze, GazeTR, GazePTR, GazeSetMerge [34], and DCGaze — as well as our method, Resnet is used as the backbone network, and we load the pretrained ResNet parameters from ImageNet during training. For CrossGaze [35], the InceptionResNet pretrained on VGGFace2 is used, and for MAFM-Gaze, Kaiming initialization weights [36] are applied. The performance results of these settings are shown in Table 1.

As shown in Table 1, most models perform better using a single face image compared to a single eye image. This is expected, as the face image not only contains complete eye information but also includes head pose-related data, which provides additional context for more accurate gaze estimation.

Our algorithm excels in accuracy, achieving the lowest angular errors on the MPIIFaceGaze(4.88°), EyeDiap(6.69°) and ETH-XGaze(4.46°) datasets compared to other methods, with improvements of 18.5%, 1.0%, and 4.7% on these datasets, respectively. These results demonstrate that our approach provides superior accuracy across multiple datasets.

In terms of computational efficiency, our method also stands out with the lowest GFLOPs of 3.12, significantly lower than algorithms like DCGaze (113.49 GFLOPs) and GazeSetMerge (80.99 GFLOPs). Additionally, with a parameter size of 25.29 MB, our method is more lightweight compared to models such as GazeSetMerge (80.99 MB) and DCGaze (113.49 MB). Overall, our method offers an excellent combination of low error rates, computational efficiency, and moderate model size, making it a highly efficient and practical solution.

4.4. Ablation studies

To validate the effectiveness of each component in our method, we performed ablation studies by removing or modifying specific parts of the network and evaluating their impact on performance.

We first removed the head pose field branch, the pseudo head pose branch, and the MSF module from the network, and then gradually added these components back to evaluate their individual contributions. The ablation results are summarized in Table 2. After introducing the head pose field branch, the error on MPIIFaceGaze decreased from 7.33° to 5.76°, on EyeDiap from 7.18° to 6.91°, and on ETH-XGaze from 6.48° to 6.37°, demonstrating consistent performance gains. These results indicate that the head pose field branch effectively compensates for the lack of head pose awareness caused by the absence of normalization.

Building upon this, incorporating MSF led to further improvements in gaze estimation accuracy, highlighting its effectiveness in integrating

Table 3
Ablation study for MSF module. MLP refers to concatenating the feature vectors and then using them directly for prediction, Mean refers to averaging the feature vectors of three branches, MSF refers to the feature fusion method we propose.

Field	Head	Fusion	MPIIFaceGaze	EyeDiap	ETH-XGaze
✓	✓	MLP	5.73°	7.15°	4.91°
✓	✓	Mean	5.47°	7.06°	4.80°
✓	✓	MSF	**4.88°**	**6.69°**	**4.46°**

eye gaze features with head pose field features. When all components were included, the model achieved the best performance: the error on MPIIFaceGaze was reduced from 7.33° to 4.88°, corresponding to a 33.4% improvement; on EyeDiap, it decreased from 7.18° to 6.69°, a 6.8% improvement; and on ETH-XGaze, it dropped from 6.48° to 4.46°, yielding a 31.2% improvement.

Next, we conducted comparative experiments to validate the effectiveness of the proposed MSF fusion module. Specifically, we replaced the feature aggregation component with two alternative fusion strategies. In the first variant, features extracted by the deep encoders were concatenated and passed through fully connected layers to produce the final prediction; this approach is referred to as MLP fusion. In the second variant, features from the deep encoders were directly averaged and used to predict gaze direction, which we denote as Mean fusion. As shown in Table 3, the proposed MSF method consistently outperforms both the MLP and Mean fusion approaches, demonstrating the effectiveness and superiority of the MSF fusion module.

To further demonstrate the effectiveness of the proposed head pose branch, we reproduced the method of Zhang [37], which incorporates head pose information to assist gaze estimation. For a fair comparison, the original LeNet5 backbone used by Zhang et al. was replaced with ResNet18. The predicted head pose yaw and pitch angles were then directly concatenated with the features of the final fully connected layer to regress eye-region gaze.

In addition, during head pose estimation with the pre-trained 6DRepnet, intermediate head pose feature maps can be obtained. Leveraging these features, we designed an additional experiment in which both the 6DRepnet head pose feature maps and the eye-region features were fed into the MSF module, followed by gaze estimation from the fused features. This setting is also included as a comparative baseline, and the corresponding results are reported in Table 4.

As shown in Table 4, Zhang's method yields almost no performance improvement for unnormalized eye gaze estimation. This outcome is expected, since the head pose yaw and pitch angles are generated by the pre-trained 6DRepnet and inevitably contain estimation errors. Directly concatenating these angle values into the network can therefore introduce noise, significantly weakening the effectiveness of head pose guidance. In contrast, the fourth setting, which utilizes the intermediate feature maps from 6DRepnet, achieves a notable improvement over the baseline. This is because feature maps encode much richer and more implicit information than explicit angle values. Through iterative training, the network is able to selectively extract gaze-relevant cues from these implicit representations, thereby providing more effective guidance for eye gaze estimation.

To investigate the impact of head pose angles on gaze estimation, we used the head pose rotation vectors provided by the MPIIFaceGaze dataset to obtain the pitch, yaw, and roll angles of the head. We then calculated the angle estimation errors between these and the angles estimated by 6DRepnet. The errors between 6DRepnet and the dataset—provided head pose were 12.62° for pitch, 8.77° for yaw, and 1.20° for roll, with an average error of 7.13°. Subsequently, we replaced the pseudo head pose angles from 6DRepnet with the dataset-provided head pose angles for supervised training of the model. The results are shown in Table 5.

The model trained with the dataset-provided head pose angles had an error of 5.29° on MPIIFaceGaze, which is 0.41° higher than that of 6DRepnet. There could be several reasons for this. One possibility is that the rotation vectors provided by the dataset might not be very accurate. Another reason is the alignment of head pose and gaze features: the pseudo head pose generated by 6DRepnet is inferred from the face, which is also an input to the RealGaze model. This alignment between the head pose features and gaze features likely helps the model leverage both cues more effectively during training. This shared input may improve the network's ability to understand and predict gaze, as the head pose information is more intrinsically linked to the face features that are already being processed.

4.5. Integration of our framework into other methods

To validate the generalizability of our algorithmic framework, we evaluated several existing open-source gaze estimation methods, including Gaze360 [9], XGaze [8], GazeTR [10], and GazePTR [11]. We first performed gaze estimation using a single real eye image. Subsequently, we replaced the eye gaze estimation backbone in our framework with each of these algorithms and trained the resulting models using real eye image together with the head pose field and real face image.

When measuring the model inference time, we set the batch size to 16 and use a single NVIDIA RTX 3090 GPU. The inference is repeated 300 times to compute the average time, reported in milliseconds. For normalization, we tested the normalization time for a single image on an Intel(R) Xeon(R) Gold 6342 CPU. After repeating the process 300 times and taking the average, the normalization time for a single image was found to be 7.56 ms.

As reported in Table 6, when our framework is integrated into existing algorithms, the gaze estimation performance shows significant improvement compared to Real Eye, and is only slightly lower than Norm Eye. Additionally, the inference time is also similar to that of Norm Eye. This indicates that our method can effectively replace the normalization approach, with minimal reduction in both performance and computational speed.

4.6. Cross camera experiments

To further demonstrate the superiority of our proposed method, we conducted cross-camera experiments. Since MPIIFaceGaze and EyeDiap datasets are captured using only one or two cameras, they lack sufficient cross-camera data. Therefore, we chose the ETH-XGaze dataset for this experiment. ETH-XGaze includes data captured from 18 cameras. For this study, we selected three camera pairs: camera0 and camera1, camera2 and camera3, and camera4 and camera5. We trained our model on the dataset captured by one camera in each pair and tested it on the dataset captured by the other camera. Considering that different head pose estimation algorithms might influence the results, we used not only 6DRepnet but also the latest open-source algorithm, Opal [38]. The results are shown in Table 7.

As shown in Table 7, our method exhibits strong performance in cross-camera settings. Compared with the normalized eye approach, it achieves a substantial improvement, and it also significantly outperforms the baseline using real eye images. Although slight differences can be observed between the results obtained with 6DRepnet and Opal, the variations are minimal and do not affect the overall conclusions. This suggests that while different head pose estimation algorithms may introduce minor variations in performance, their impact is not decisive.

Table 4
Comparison with existing head pose-assisted gaze estimation methods and 6DRepnet feature map fusion.

Algorithms	Input	MPIIFaceGaze	EyeDiap	ETH-XGaze
Baseline(Resnet18)	Eye	7.33°	7.18°	6.48°
Zhang [37]	Eye+Head Pose Angle	7.33°	7.16°	6.45°
6DRepnet feature map	Eye+Face	6.58°	7.06°	5.07°
Ours	Eye+Face+Field	**4.88°**	**6.69°**	**4.46°**

Table 5
Comparison with different head pose supervisions on MPIIFaceGaze.

Algorithms	Input	Head Pose supervision	MPIIFaceGaze
RealGaze	Eye+Face+Field	Provided by Dataset	5.29°
		6DRepnet [5]	4.88°

4.7. Comparison of normalized and real image results on EyeDiap

Cameras are typically shipped with factory-calibrated intrinsic parameters, such as focal length and principal point. However, variations in installation and operating conditions often cause the actual intrinsic parameters after deployment to deviate from these factory settings, making additional camera calibration necessary to recover the true parameters. To compare our method with normalization-based approaches, we first normalized the images using the true intrinsic parameters to obtain correctly normalized data. We then performed normalization using the factory-calibrated intrinsic parameters. Since the focal length usually remains unchanged, we introduced offsets to the principal point based on the factory parameters, thereby generating intrinsic parameters with different levels of error.

Specifically, for the EyeDiap dataset, the true focal lengths are $fx = 522.33$ and $fy = 522.35$. We set the factory-calibrated focal lengths to $fx = fy = 520$ and applied principal point offsets of 0, ±1%, and ±2%. For example, for the EyeDiap dataset with raw images of width 640 and height 480, a uv bias of 0 corresponds to $u = 320$ and $v = 240$, while a uv bias of ±2% corresponds to $u = 326$ and $v = 244$. We trained and tested the baseline network on both normalized and real images, and we also trained and tested the RealGaze network on real images. The results, shown in Table 8, illustrate the performance differences under various intrinsic parameter settings.

As shown in Table 8, the gaze estimation error of normalization-based methods is highly sensitive to inaccuracies in the camera principal point. Compared with normalization using the true intrinsic parameters, the error increases dramatically when factory-calibrated parameters are used, reaching up to 9.28°. This is because the true intrinsic parameters can differ substantially from the factory settings, and relying on factory-calibrated values introduces significant normalization errors. When a bias is applied to the principal point of the factory parameters, the gaze estimation error is partially reduced. For example, with a principal point offset of ±2%, the error decreases to 7.87°, indicating some improvement over the factory-calibrated case. However, this error remains considerably higher than the 6.44° achieved using the true intrinsic parameters, and is even worse than the performance obtained directly from unnormalized real images.

In contrast, the proposed RealGaze method consistently improves gaze estimation on real images, reducing the error from 7.18° to 6.69°, which is close to the result obtained with true intrinsic parameters. These results demonstrate that RealGaze is largely unaffected by camera intrinsic parameters, offering strong adaptability and superior performance in practical deployment scenarios.

4.8. Feature map visualization

To better illustrate the effectiveness of the MSF module, we visualize the feature maps before fusion. The network contains three branches: the eye branch, the head branch, and the pose field branch, each producing a shallow feature map through its corresponding encoder. The head feature map and the pose field feature map are first fused using the MSF module, after which the resulting representation is further fused with the eye feature map to generate the final fused feature map.

As shown in Fig. 5, head _feat represents the head feature map, field _feat represents the field feature map, and eye _feat represents the eye feature map, with other feature maps being similarly defined. The first two rows in the results come from the MPIIFaceGaze dataset, while the last two rows are from the ETH-XGaze dataset.

As observed in the figure, the feature map produced solely by the shallow head encoder appears relatively disorganized, with ambiguous and weakly structured representations. After being fused with the pose field feature map through the MSF module, the activation in the central facial region is significantly enhanced, while responses in gaze-irrelevant areas, such as peripheral regions, are noticeably suppressed. With further guidance from the eye feature map, the final fused feature map concentrates predominantly on the eye region, assigning even higher weights to areas directly related to gaze. This progression clearly demonstrates the effectiveness of the MSF-based feature fusion: gaze-irrelevant regions are progressively filtered out, and greater emphasis is placed on the eye region, which is most critical for accurate gaze estimation.

It is worth noting that, since the head pose field branch is also involved in the overall training of the model, the parameters of this branch are influenced by the other two branches. Therefore, field _feat does not always correspond one-to-one with the head pose field. This is why we do not directly multiply the head pose field by eye _feat or head _feat, but instead treat it as a separate branch for parameter updates.

4.9. Qualitative results

We randomly selected samples from the three datasets and visualized the qualitative results of RealGaze and the baseline, as shown in Fig. 6. In the figure, red lines denote the predictions of RealGaze, blue lines represent the baseline results, and green lines indicate the ground truth. The first row corresponds to MPIIFaceGaze, the second row to EyeDiap, and the third row to ETH-XGaze. It can be observed that, for both clear and blurry images, RealGaze consistently produces predictions that are closer to the ground truth than the baseline, demonstrating the superior performance of the proposed method.

4.10. Limitations

Despite its promising performance, RealGaze has several limitations that warrant further investigation.

First, the training process relies on pseudo head pose labels generated by external head pose estimation algorithms. Although experiments show that different head pose estimators lead to similar performance trends, the pseudo labels inevitably contain estimation errors, particularly under extreme poses, occlusions, or low image quality. These errors may propagate into the gaze estimation network and limit its performance in challenging scenarios. Future work could explore jointly learning head pose and gaze estimation in a unified end-to-end framework to reduce reliance on external models.

Second, the method still depends on reliable eye keypoint detection for cropping the eye region. While this requirement does not involve camera calibration, inaccurate keypoint localization may result

Table 6
Integration of our framework into other methods. Norm Eye represents a single normalized eye image, Real Eye represents a single real eye image, and Real Eye+Face+Field represents a combination of a single real eye image, a single real face image, and a single head pose field image. The time taken to normalize a single raw image is 7.56 ms. Therefore, for Norm Eye, we included the normalization time when comparing the inference time across different methods.

Algorithms	Years	Input	MPIIFaceGaze	EyeDiap	Inference (ms)
Gaze360 [9]	ICCV2019	Norm Eye	4.66°	6.74°	11.65(4.09 + 7.56)
		Real Eye	7.54°	7.32°	4.09
Gaze360+Field+Head+MSF		Real Eye+Face+Field	5.23° ↓ 30.64%	7.00° ↓ 4.37%	12.20
XGaze [8]	ECCV2020	Norm Eye	4.48°	6.33°	17.56(10.02 + 7.56)
		Real Eye	7.34°	6.91°	10.02
XGaze+Field+Head+MSF		Real Eye+Face+Field	4.82° ↓ 34.33%	6.61° ↓ 4.34%	17.72
GazeTR [10]	ICPR2022	Norm Eye	4.52°	6.69°	15.84(8.28 + 7.56)
		Real Eye	7.36°	7.14°	8.28
GazeTR+Field+Head+MSF		Real Eye+Face+Field	5.12° ↓ 30.43%	6.75° ↓ 5.46%	17.41
GazePTR [33]	CVPR2024	Norm Eye	4.74°	6.34°	17.76(10.20 + 7.56)
		Real Eye	7.36°	7.00°	10.20
GazePTR+Field+Head+MSF		Real Eye+Face+Field	5.15° ↓ 30.03%	6.91° ↓ 1.29%	19.52

Table 7
Cross camera results. Cam0→1 denotes training on the sub-training set of 56 subjects from camera0 and testing on the sub-test set of 12 subjects from camera1. 6d represents the use of the 6DRepnet algorithm, while op represents the use of the Opal algorithm.

Image type	Algorithm	Cam0→1	Cam1→0	Cam2→3	Cam3→2	Cam4→5	Cam5→4
Norm Eye	Baseline	5.93°	8.00°	7.24°	7.44°	6.69°	6.09°
Real Eye	Baseline	6.14°	7.47°	4.98°	5.44°	6.83°	6.22°
	Ours(6d)	4.99°	**5.89°**	**4.27°**	**5.22°**	5.36°	**4.57°**
	Ours(op)	**5.51°**	6.44°	4.55°	5.40°	**5.87°**	5.05°

Table 8
Performance comparison of normalized and real images. The standard focal length of EyeDiap is $fx = fy = 520$. Bold indicates the best result, and underline indicates the second-best result.

Image type			Algorithm	EyeDiap
Norm Eye	True Fx Fy U V		Baseline	**6.44°**
	Standard Fx Fy	Uv Bias = 2%		7.87°
		Uv Bias = 1%		8.23°
		Uv Bias = 0		9.28°
		Uv Bias = −1%		10.70°
		Uv Bias = −2%		11.09°
Real Eye			Baseline	7.18°
			Ours	<u>6.69°</u>

in suboptimal eye crops, which can negatively affect gaze prediction. Improving robustness to keypoint detection errors or developing keypoint-free eye region extraction strategies would further enhance the applicability of the method.

Third, there is a trade-off between accuracy and computational complexity. Compared to lightweight models such as Gaze360, RealGaze introduces additional branches and fusion modules, leading to increased computational cost. Although the current efficiency is sufficient for many real-time or near-real-time applications, further optimization through model compression, lightweight fusion designs, or knowledge distillation could be beneficial for deployment on resource-constrained devices.

Finally, the current study focuses on single-eye gaze estimation. While the results demonstrate that a single eye combined with head pose information can achieve competitive or superior performance, incorporating binocular cues or extending the framework to multi-eye or multi-person settings may further improve robustness in complex interaction scenarios.

5. Conclusion and future work

In this work, we propose RealGaze, an end-to-end gaze estimation framework that directly processes real, unnormalized eye images, offering a solution to the limitations of traditional normalization-based methods. By leveraging a novel head pose field branch and a pseudo head pose strategy, RealGaze effectively incorporates head pose information without the need for image normalization, enabling accurate gaze prediction across various real-world scenarios.

Our extensive experiments across multiple datasets, including MPIIFaceGaze, EyeDiap, and ETH-XGaze, demonstrate that RealGaze outperforms existing methods, achieving state-of-the-art results with significant improvements in gaze estimation accuracy. Furthermore, RealGaze shows robust performance under cross-camera conditions and without reliance on precise camera intrinsics, making it highly adaptable to practical applications where camera calibration is challenging.

The Multi-Scope Fusion (MSF) module introduced in our approach effectively integrates diverse feature sources, enhancing model performance and ensuring accurate gaze predictions even in the presence of head pose variations. Through this work, we highlight the importance of considering head pose directly within gaze estimation networks, marking a shift towards more flexible, real-world gaze tracking solutions.

Future work will focus on several key areas to extend and improve RealGaze's capabilities. First, we aim to optimize the framework for multi-eye and multi-person gaze estimation, which would allow for more complex scenarios such as group interactions or environments with multiple subjects. Second, further research will explore real-time, on-device deployment, optimizing the model for faster inference without compromising accuracy, which is crucial for applications in mobile devices and edge computing. We also plan to integrate additional modalities, such as facial expressions or head movement, to enhance gaze estimation accuracy and robustness in more dynamic environments.

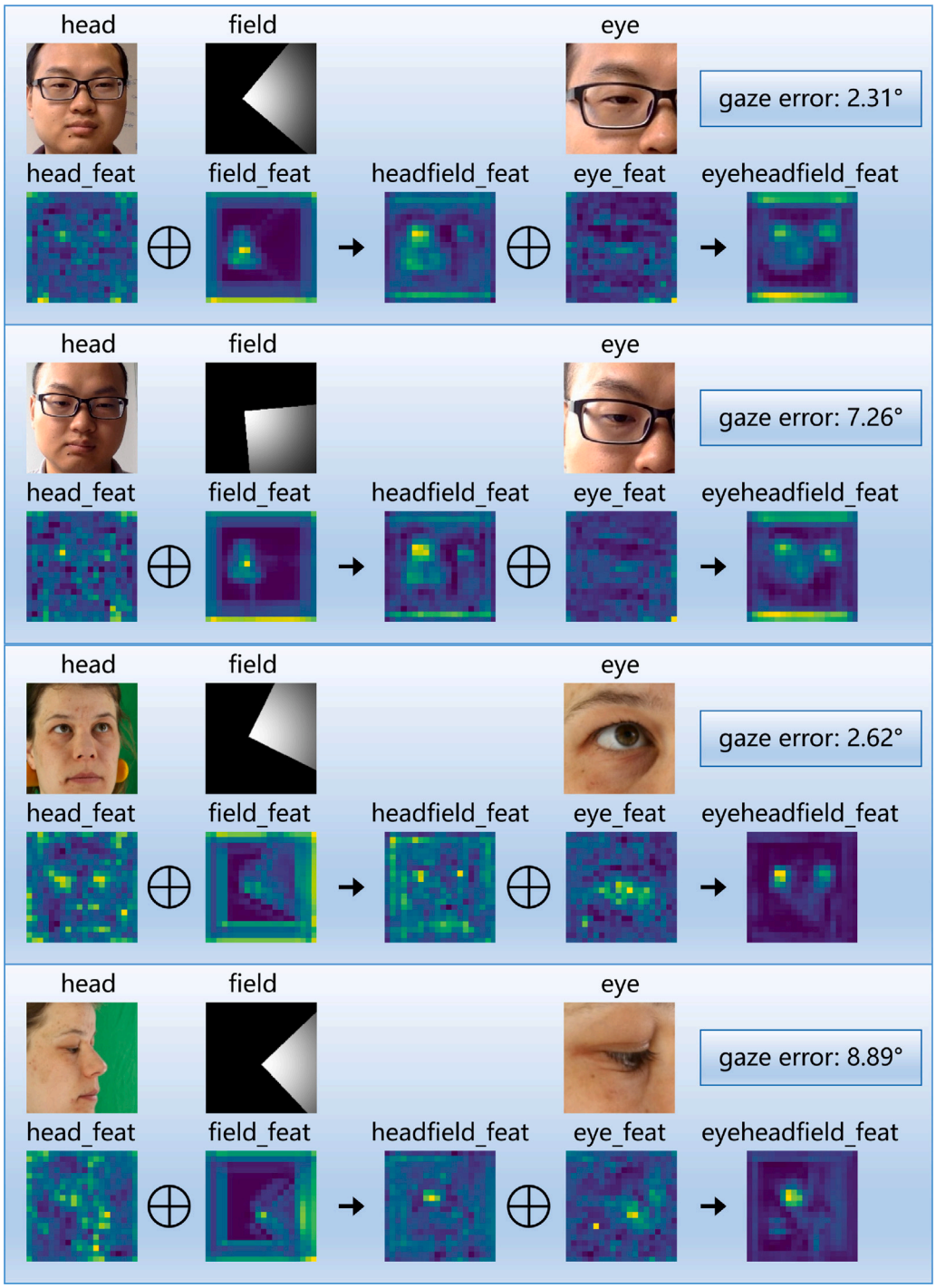

Fig. 5. Feature map visualization.

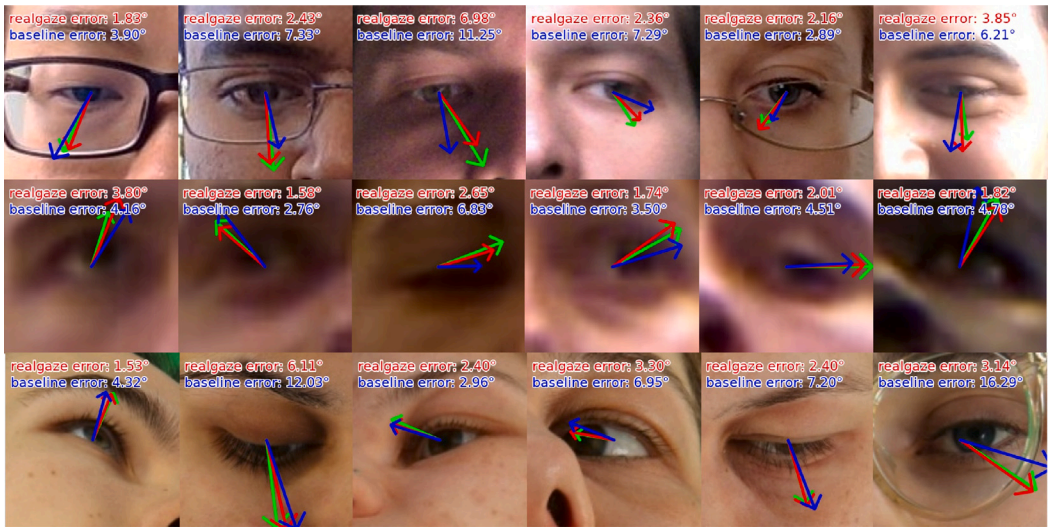

Fig. 6. Qualitative results. The first row shows MPIIFaceGaze, the second row shows EyeDiap, and the last row shows ETH-XGaze. Red arrows indicate RealGaze, blue arrows indicate baseline, and green arrows indicate the ground truth of gaze.

Another avenue for future development is the exploration of joint training for gaze and head pose estimation, reducing dependency on external head pose estimation models and allowing for more accurate, self-contained gaze estimation. Finally, we aim to broaden the application scope by evaluating RealGaze in specialized domains, such as driver assistance systems, virtual reality, and assistive technologies for individuals with disabilities, where gaze tracking could provide significant value.

By addressing these challenges and expanding the framework's scope, we believe RealGaze can evolve into a versatile, high-performance gaze estimation tool with widespread applicability in various real-world scenarios.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] P.R. Krishnappa Babu, U. Lahiri, Understanding the role of proximity and eye gaze in human–computer interaction for individuals with autism, J. Ambient. Intell. Humaniz. Comput. (2024) 1–15.

[2] B. Cebeci, M.B. Askin, T.K. Capin, U. Celikcan, Gaze-directed and saliency-guided approaches of stereo camera control in interactive virtual reality, Comput. Graph. (2024).

[3] P.K. Sharma, P. Chakraborty, A review of driver gaze estimation and application in gaze behavior understanding, Eng. Appl. Artif. Intell. 133 (2024) 108117.

[4] B.-W. Chu, W.-L. Ou, R.C.-H. Chang, C.-P. Fan, Calibration-free gaze estimation by combination with hand and facial features detection for interactive advertising display, in: 2024 IEEE International Conference on Consumer Electronics, ICCE, IEEE, 2024, pp. 1–3.

[5] T. Hempel, A.A. Abdelrahman, A. Al-Hamadi, Toward robust and unconstrained full range of rotation head pose estimation, IEEE Trans. Image Process. 33 (2024) 2377–2387.

[6] X. Zhang, Y. Sugano, M. Fritz, A. Bulling, It's written all over your face: Full-face appearance-based gaze estimation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2017, pp. 51–60.

[7] K.A. Funes Mora, F. Monay, J.-M. Odobez, Eyediap: A database for the development and evaluation of gaze estimation algorithms from rgb and rgb-d cameras, in: Proceedings of the Symposium on Eye Tracking Research and Applications, 2014, pp. 255–258.

[8] X. Zhang, S. Park, T. Beeler, D. Bradley, S. Tang, O. Hilliges, Eth-xgaze: A large scale dataset for gaze estimation under extreme head pose and gaze variation, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part V 16, Springer, 2020, pp. 365–381.

[9] P. Kellnhofer, A. Recasens, S. Stent, W. Matusik, A. Torralba, Gaze360: Physically unconstrained gaze estimation in the wild, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6912–6921.

[10] Y. Cheng, F. Lu, Gaze estimation using transformer, in: 2022 26th International Conference on Pattern Recognition, ICPR, IEEE, 2022, pp. 3341–3347.

[11] Y. Cheng, Y. Zhu, Z. Wang, H. Hao, Y. Liu, S. Cheng, X. Wang, H.J. Chang, What do you see in vehicle? Comprehensive vision solution for in-vehicle gaze estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 1556–1565.

[12] X. Zhao, Y. Huang, Y. Tian, M. Tian, Episode-based personalization network for gaze estimation without calibration, Neurocomputing 513 (2022) 36–45.

[13] Q. Wang, H. Wang, R.-C. Dang, G.-P. Zhu, H.-F. Pi, F. Shic, B.-l. Hu, Style transformed synthetic images for real world gaze estimation by using residual neural network with embedded personal identities, Appl. Intell. 53 (2) (2023) 2026–2041.

[14] A.A. Abdelrahman, T. Hempel, A. Khalifa, A. Al-Hamadi, Fine-grained gaze estimation based on the combination of regression and classification losses, Appl. Intell. 54 (21) (2024) 10982–10994.

[15] Y. Jia, Z. Liu, Y. Lv, X. Lu, X. Liu, J. Chen, Frequency-spatial interaction network for gaze estimation, Displays 86 (2025) 102878.

[16] G. Zhang, Y. Wang, R. Yan, X. Fu, MAFM-Gaze: Multi-scale adaptive feature modulation for in-vehicle gaze estimation, Displays (2025) 103226.

[17] L. Zhang, Y. Tian, X. Wang, W. Xu, Y. Jin, Y. Huang, Differential contrastive training for gaze estimation, in: Proceedings of the 33rd ACM International Conference on Multimedia, 2025, pp. 3477–3486.

[18] X. Zhang, Y. Sugano, A. Bulling, Revisiting data normalization for appearance-based gaze estimation, in: Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications, 2018, pp. 1–9.

[19] Q. Huang, A. Veeraraghavan, A. Sabharwal, Tabletgaze: dataset and analysis for unconstrained appearance-based gaze estimation in mobile tablets, Mach. Vis. Appl. 28 (2017) 445–461.

[20] Y. Sugano, Y. Matsushita, Y. Sato, Learning-by-synthesis for appearance-based 3d gaze estimation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 1821–1828.

[21] H. Balim, S. Park, X. Wang, X. Zhang, O. Hilliges, Efe: End-to-end frame-to-gaze estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 2688–2697.

[22] M. Zhang, Y. Liu, F. Lu, Gazeonce: Real-time multi-person gaze estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 4197–4206.
[23] H. Huang, M. Zhang, J. Yang, Z. Li, J. Li, Y. Guo, MAGE: A multi-task architecture for gaze estimation with an efficient calibration module, 2025, arXiv preprint arXiv:2505.16384.
[24] Y. Bao, Y. Cheng, Y. Liu, F. Lu, Adaptive feature fusion network for gaze tracking in mobile tablets, in: 2020 25th International Conference on Pattern Recognition, ICPR, IEEE, 2021, pp. 9936–9943.
[25] J. Gideon, S. Su, S. Stent, Unsupervised multi-view gaze representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 5001–5009.
[26] J.-S. Yun, Y. Na, H.H. Kim, H.-I. Kim, S.B. Yoo, HAZE-Net: High-frequency attentive super-resolved gaze estimation in low-resolution face images, in: Proceedings of the Asian Conference on Computer Vision, 2022, pp. 3361–3378.
[27] Y. Hisadome, T. Wu, J. Qin, Y. Sugano, Rotation-constrained cross-view feature fusion for multi-view appearance-based gaze estimation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 5985–5994.
[28] H. Chen, H. Liu, S. Lan, W. Wang, Y. Qiao, Y. Li, G. Deng, DMAGaze: Gaze estimation based on feature disentanglement and multi-scale attention, 2025, arXiv preprint arXiv:2504.11160.
[29] C. Li, E. Tong, K. Zhang, N. Cheng, Z. Lai, Z. Pan, Gaze estimation based on a multi-stream adaptive feature fusion network, Appl. Sci. 15 (7) (2025) 3684.
[30] Z. Wu, Y. Lin, H. Cheng, C. Kong, W. Zhou, H. Li, Mixgaze: a dually supervised mixed attention network for gaze estimation, Multimedia Syst. 32 (1) (2026) 15.
[31] D.E. King, Dlib-ml: A machine learning toolkit, J. Mach. Learn. Res. 10 (2009) 1755–1758.
[32] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
[33] Y. Cheng, H. Wang, Y. Bao, F. Lu, Appearance-based gaze estimation with deep learning: A review and benchmark, IEEE Trans. Pattern Anal. Mach. Intell. (2024).
[34] L. Wu, B.E. Shi, Merging multiple datasets for improved appearance-based gaze estimation, in: International Conference on Pattern Recognition, Springer, 2024, pp. 77–90.
[35] A. Catruna, A. Cosma, E. Radoi, Crossgaze: A strong method for 3d gaze estimation in the wild, in: 2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition, FG, IEEE, 2024, pp. 1–5.
[36] K. He, X. Zhang, S. Ren, J. Sun, Delving deep into rectifiers: Surpassing human-level performance on imagenet classification, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 1026–1034.
[37] X. Zhang, Y. Sugano, M. Fritz, A. Bulling, Appearance-based gaze estimation in the wild, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 4511–4520.
[38] A. Cobo, R. Valle, J.M. Buenaposada, L. Baumela, On the representation and methodology for wide and short range head pose estimation, Pattern Recognit. 149 (2024) 110263.